

23 Nuclear physics

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Nuclear mass defect is the difference between separate constituent masses and nuclear mass; binding energy is Δmc^2 . Activity is decay events per second, $A = \lambda N$. Half-life is the time for half the undecayed nuclei to remain; decay is random for one nucleus but predictable statistically for a large sample.

Worked example: follow the reasoning

An 800 Bq source with half-life 3 h has 400 Bq after 3 h and 100 Bq after 9 h, excluding background.

Quick check · multiple choice

A What is one becquerel?

[1]

- A One joule per second
- B One nucleus per kilogram
- C One decay per second
- D One half-life per second

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A sample starts at 640 Bq and its half-life is 5.0 h. Find activity after 15 h, then state what one becquerel means.

[3]

Final answer: _____

Answers and explanation

Quick check: C. One decay per second

Question 1 [3 marks]

15 h is three half-lives [1]. $A = 640/2^3 = 80$ Bq [1]. One Bq is one nuclear decay per second [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.